

Amaan Khan

(773) 993-6396 | mkhan226@asu.edu | linkedin.com/in/amaankhan2000/ | github.com/AmaanKhan2000 | amaankhan.dev

EXPERIENCE

Software Engineer II

July 2022 – August 2024

Verizon

- Improved platform reliability by **30%** and cut release regressions by 20% for **6,300+** users, by engineering fault-isolated Python and **Java** backend services across 3+ high-availability microservices with end-to-end CI/CD coverage.
- Slashed P99 query latency by **72%** on Verizon's **50M+** customer account record system, measured under peak subscriber load, by deploying a **Redis** cache with TTL eviction across high-frequency lookup endpoints.
- Streamlined post-deployment incidents by **25%** and MTTR by 20% across **3,000+** network and billing microservices, by executing phased migration with **Kubernetes** rolling deployments, canary gates, and sprint-level checkpoints.
- Freed **28 hrs/week** of manual reporting overhead for Verizon's NOC teams, providing near-real-time service visibility, by building **TypeScript**/Angular reactive dashboards on live microservice REST API feeds.
- Accelerated Verizon's telecom microservice delivery by **33%**, measured by deployment frequency from sprint to production, by automating **GitLab CI/CD** pipelines with carrier-grade quality gates for builds and containerization.
- Curbed inter-service latency across billing, provisioning, and notification services by **55%**, measured at peak subscriber load, by implementing **Apache Kafka** event streaming to replace synchronous REST call chains.

AI Full Stack Engineer

June 2021 – September 2021

Banao Technologies

- Designed an ML content ranking pipeline in **Java** and **Python**, replacing legacy rule-based systems and cutting manual content curation by **70%** through structured A/B experimentation.
- Developed **REST** and **GraphQL** APIs serving personalized content ranking predictions at **120ms** average latency, enabling product teams to consume real-time recommendations without direct engineering intervention.
- Engineered **Python** preprocessing pipelines for **50K+** daily content records, applying OOP design to build modular components that standardized feature extraction for training and serving the ranking model.
- Delivered **full-stack** TypeScript and **React** features for customer-facing interfaces integrated with ML ranking APIs, enabling **10+** product stakeholders to monitor real-time predictions and A/B results independently.

Software Developer

March 2021 – June 2021

Vidya (Formerly Across The Globe)

- Enhanced learning analytics dashboards in **Python** and **PostgreSQL**, enabling data-driven decisions for **3+** academic teams through optimized SQL queries and REST API integrations.
- Refactored **PostgreSQL** queries via compound indexing and JOIN optimization, cutting report generation time by **60%** and enabling teams to surface learning outcome data on demand.
- Designed and documented **8+ REST API** endpoints in Python that expose learning data to partner institutions, enabling self-serve integrations and reducing custom development overhead per client.
- Collaborated with curriculum teams to translate learning outcome KPIs into **SQL** queries and **Python** pipelines, replacing manual reporting workflows and saving **20+** hours per week.

TECHNICAL SKILLS

Languages & Frameworks: Java, Python, Go, TypeScript, JavaScript (ES6+), C++, Node.js, Spring Boot, Angular, React, Flask
AI & Machine Learning: PyTorch, TensorFlow, scikit-learn, LangChain, OpenAI API, LLMs, RAG, NLP, Prompt Engineering
Databases & Cloud: PostgreSQL, MongoDB, Redis, DynamoDB, Kafka, Neo4j, Airflow, AWS (Lambda, EC2, S3, SQS, SNS, CloudWatch), GCP, Azure, Docker, Kubernetes, Terraform
Tools & DevOps: GitLab CI/CD, Prometheus, Grafana, Git, JUnit, REST APIs, GraphQL, gRPC, Claude Code, GitHub Copilot
Methodologies: Agile, Test-Driven Development (TDD), Integration Testing, ETL Pipelines, CI/CD

PROJECTS

RepMem: Representation-Adaptive Memory Routing for Long-Term LLMs | *Python, Claude Code, Prompt Engineering* Jan 2026

- Designed a **5-format** parallel memory system routing conversations across atomic facts, **KG triples**, session summaries, temporal events, and contradiction logs for query-adaptive LLM recall.
- Outperformed Mem0, Zep, A-MEM, and FluxMem by **1.6×**, with the learned **Gradient Boosting** router achieving 0.838 LLM-as-Judge score within 0.012 of Oracle ceiling on LongMemEval.
- Achieved **22×** token efficiency over Full Context while validating diagonal format-query alignment across LongMemEval and LoCoMo, confirming generalization across all four query types.

Distributed API Test Automation for Containerized Services | *Java, Python, Docker* August 2025 – December 2025

- Built a multi-threaded distributed test runner in **Java** parallelizing REST API validation across **Docker** containers, cutting execution time by 20% and catching regressions pre-deployment.
- Enforced contract-driven API validation as a hard **CI/CD** gate, applying unit and integration tests with deterministic data seeding to ensure high code quality.

EDUCATION

Arizona State University

Master of Science in Data Science and Analytics | Teaching Assistant

Tempe, AZ

Aug. 2024 – May 2026

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science, Specialization in AI and ML | GPA: 3.7/4.0

July 2018 – May 2022